MINIO
OBJECT STORE
AGPLV3 LICENSE

User

- Object Browser
- Access Keys
- Documentation

Administrator

- Buckets
- Policies
- Identity
- Monitoring
- Events
- Tiering
- Site Replication
- Configuration

← Object Browser

Start typing to filter objects in the bucket

analytics

Created on: **Fri, Feb 27 2026 00:01:13 (GMT+4)** Access: **CUSTOM** 2.2 KiB - 1 Object

Rewind | Refresh | Upload

analytics

Create new path

Name	Last Modified	Size
analytic_result_2026_02.csv	Today, 15:18	2.2 KiB

Структура CSV (через mc)

```
(.venv) PS D:\Course-DE-fall_2025\retail_project> docker exec -it retail_minio_mc sh -lc "mc cat local/analytics/analytic_result_2026_02.csv | head"
"customer_id","new_customer","loyal_customer","recurrent_buyer","delivery_user","prefers_cash","prefers_card","weekend_shopper","weekday_shopper","morning_shopper","night_shopper",
"store_loyal","switching_store","recent_high_spender","low_cost_buyer","sum_7d","sum_30d"
"cus-1000",0,1,0,0,0,0,0,0,0,0,0,0,1,0,0
"cus-1001",0,1,0,1,1,0,0,1,0,0,0,0,0,1,0,0
"cus-1002",0,1,1,0,1,0,0,1,0,0,0,0,0,1,0,0
"cus-1003",1,1,1,0,0,0,0,1,0,0,0,1,0,0,0,1980
"cus-1004",0,1,1,1,0,0,0,1,1,0,0,1,0,0,0,1350
"cus-1005",0,1,1,0,0,0,0,1,0,0,0,0,0,1,0,0
"cus-1006",0,1,1,1,0,1,0,1,0,1,1,0,0,0,0,920
"cus-1007",0,1,0,0,1,0,0,1,1,0,0,0,0,1,0,0
"cus-1008",0,1,1,1,0,0,0,1,1,0,0,1,0,0,0,1420
(.venv) PS D:\Course-DE-fall_2025\retail_project>
```

Выгрузка из ClickHouse

```
INSERT INTO FUNCTION s3(
 'http://192.168.96.2:9000/analytics/analytic_result_2026_02.csv',
 'minioadmin',
 'minioadmin',
 'CSVWithNames'
)
SELECT
  customer_id,
  new_customer,
  loyal_customer,
  recurrent_buyer,
  delivery_user,
  prefers_cash,
  prefers_card,
  weekend_shopper,
  weekday_shopper,
  morning_shopper,
  night_shopper,
  store_loyal,
  switching_store,
  recent_high_spender,
  low_cost_buyer,
  sum_7d,
  sum_30d
FROM retail_analytics.customer_feature_matrix;

SET s3_truncate_on_insert = 1

Query id: e288d886-2ab8-4289-b7fd-c9f551e02eed

Ok.

0 rows in set. Elapsed: 0.001 sec.
```

```
INSERT INTO FUNCTION s3('http://192.168.96.2:9000/analytics/analytic_result_2026_02.csv', 'minioadmin', 'minioadmin', 'CSVWithNames') SELECT
    customer_id,
    new_customer,
    loyal_customer,
    recurrent_buyer,
    delivery_user,
    prefers_cash,
    prefers_card,
    weekend_shopper,
    weekday_shopper,
    morning_shopper,
    night_shopper,
    store_loyal,
    switching_store,
    recent_high_spender,
    low_cost_buyer,
    sum_7d,
    sum_30d
FROM retail_analytics.customer_feature_matrix

Query id: 5d1cf569-1299-496d-aae5-62e48189e861

Ok.

0 rows in set. Elapsed: 0.024 sec.
```

Витрина

```
SELECT *
FROM retail_analytics.customer_feature_matrix
LIMIT 5

Query id: 2b63299f-bbb7-46c5-8200-6dc06d2ae832

   ┌─customer_id─┬─new_customer─┬─loyal_customer─┬─no_purchases─┬─inactive_14_30─┬─recurrent_buyer─┬─delivery_user─┬─prefers_cash─┬─prefers_card─┬─weekend_shopper─┬─weekday_shopper
─┬─morning_shopper─┬─night_shopper─┬─store_loyal─┬─switching_store─┬─recent_high_spender─┬─low_cost_buyer─┬─sum_7d─┬─sum_30d─┬────────etl_loaded_at─┐
1. │ cus-1000    │            0 │            1 │             1 │              0 │               0 │             0 │            0 │            0 │               0 │                0
 │             0 │             0 │           0 │               0 │                  0 │              1 │      0 │       0 │ 2026-03-04 10:27:43 │
2. │ cus-1001    │            0 │            1 │             0 │              0 │               0 │             1 │            1 │            0 │               0 │                1
 │             0 │             0 │           0 │               0 │                  0 │              1 │      0 │       0 │ 2026-03-04 10:27:43 │
3. │ cus-1002    │            0 │            1 │             0 │              0 │               1 │             0 │            1 │            0 │               0 │                1
 │             0 │             0 │           0 │               0 │                  0 │              1 │      0 │       0 │ 2026-03-04 10:27:43 │
4. │ cus-1003    │            1 │            1 │             0 │              1 │               1 │             0 │            0 │            0 │               0 │                1
 │             0 │             0 │           0 │               1 │                  0 │              0 │      0 │    1980 │ 2026-03-04 10:27:43 │
5. │ cus-1004    │            0 │            1 │             0 │              0 │               1 │             1 │            0 │            0 │               0 │                1
 │             1 │             0 │           0 │               1 │                  0 │              0 │      0 │    1350 │ 2026-03-04 10:27:43 │
 └─────────────┴──────────────┴────────────┴──────────────┴────────────────┴─────────────────┴───────────────┴──────────────┴──────────────┴─────────────────┴─────────────────
─┴─────────────────┴───────────────┴─────────────┴─────────────────┴─────────────────────┴────────────────┴────────┴─────────┴─────────────────────┘

5 rows in set. Elapsed: 0.006 sec.
```

ClickHouse читает файл обратно из S3

```
18b7bcbb52f7 :) SELECT *
FROM s3(
 'http://192.168.96.2:9000/analytics/analytic_result_2026_02.csv',
 'minioadmin',
 'minioadmin',
 'CSVWithNames'
)
LIMIT 5;

SELECT *
FROM s3('http://192.168.96.2:9000/analytics/analytic_result_2026_02.csv', 'minioadmin', 'minioadmin', 'CSVWithNames')
LIMIT 5

Query id: f1de5f32-1d77-4412-a919-4a90f0b286a5

   ┌─customer_id─┬─new_customer─┬─loyal_customer─┬─recurrent_buyer─┬─delivery_user─┬─prefers_cash─┬─prefers_card─┬─weekend_shopper─┬─weekday_shopper─┬─morning_shopper─┬─night_shopp
er─┬─store_loyal─┬─switching_store─┬─recent_high_spender─┬─low_cost_buyer─┬─sum_7d─┬─sum_30d─┐
1. │ cus-1000    │             0 │              1 │               0 │             0 │            0 │            0 │               0 │               0 │               0 │
 0 │           0 │               0 │                   0 │              1 │      0 │       0 │
2. │ cus-1001    │             0 │              1 │               0 │             1 │            1 │            0 │               0 │               1 │               0 │
 0 │           0 │               0 │                   0 │              1 │      0 │       0 │
3. │ cus-1002    │             0 │              1 │               1 │             0 │            1 │            0 │               0 │               1 │               0 │
 0 │           0 │               0 │                   0 │              1 │      0 │       0 │
4. │ cus-1003    │             1 │              1 │               1 │             0 │            0 │            0 │               0 │               1 │               0 │
 0 │           0 │               1 │                   0 │              0 │      0 │    1980 │
   └─customer_id─┴─new_customer─┴─loyal_customer─┴─recurrent_buyer─┴─delivery_user─┴─prefers_cash─┴─prefers_card─┴─weekend_shopper─┴─weekday_shopper─┴─morning_shopper─┴─night_shopp
er─┴─store_loyal─┴─switching_store─┴─recent_high_spender─┴─low_cost_buyer─┴─sum_7d─┴─sum_30d─┘
5. │ cus-1004    │             0 │              1 │               1 │             1 │            0 │            0 │               0 │               1 │               1 │
 0 │           0 │               1 │                   0 │              0 │      0 │    1350 │
   └─customer_id─┴─new_customer─┴─loyal_customer─┴─recurrent_buyer─┴─delivery_user─┴─prefers_cash─┴─prefers_card─┴─weekend_shopper─┴─weekday_shopper─┴─morning_shopper─┴─night_shopp
er─┴─store_loyal─┴─switching_store─┴─recent_high_spender─┴─low_cost_buyer─┴─sum_7d─┴─sum_30d─┘

5 rows in set. Elapsed: 0.022 sec.
```